

Coffee Shop Sales Analysis

1. Overview:

For this project, a thorough examination of coffee shop sales data was conducted in order to determine which products were profitable and which were not, as well as to investigate potential methods for boosting profit margins and reducing losses. The data was processed, analyzed, and visualized using Python libraries like pandas, matplotlib, and seaborn.

2. Objective:

This project's primary goals were to:

- Examine and comprehend the features and organization of the coffee shop sales dataset.

- Clean up and get the data ready for analysis, taking care of duplicates, missing values, and data type conversion.
- Examine sales performance for a variety of products, taking into account total sales, quantity sold, and profit.
- Determine which products are the most and least profitable.
- To effectively convey insights, visualize the most important findings.
- Based on the analysis, we make suggestions for boosting profit margins and reducing losses.

- If appropriate and practical given the data at hand, investigate time series analysis for profit prediction.

3. Methodology:

The project used a methodical approach:

Data Loading and Exploration: A pandas DataFrame was used to load the sales data. The purpose of the initial investigation was to determine the data types, comprehend the structure of the data, look for duplicates and missing values, and obtain descriptive statistics.

Data Preparation and Cleaning: The relevant datetime objects were created from the "transaction_date" and "transaction_time" columns. Although none were discovered in this dataset, duplicate rows were located and dealt with. To

make additional analysis easier, new columns for "sale_price," "day_of_week," and "month" were made.

Profit/Loss Analysis: The cost and profit for every transaction were calculated using a fixed cost percentage. The most and least profitable products were then determined by ranking them according to their total profit.

Data Visualization: To show the total sales and profit for the top 20 most lucrative products, bar plots were created. The distribution of profit margins across all transactions was shown using a histogram.

Profit Increase Prediction (Time Series Analysis): A selection of the most and least profitable products' daily profit data was combined. In order to determine trends or seasonality and investigate the

possibility of profit prediction, ARIMA models were tried to fit the time series data for these products.

Suggestions: Recommendations for raising profit margins and reducing losses were developed based on the profitability analysis and the information gleaned from the time series modeling attempts.

4. Challenges:

The attempt to fit ARIMA models for profit prediction was one of the biggest obstacles faced. Even though the models were fitted, more research into the model diagnostics, the time series' stationarity, and possibly other time series models or outside variables that could affect sales and profit would be necessary to interpret the results and use them for accurate forecasting. Capturing long-term trends or strong seasonal patterns is further limited

by the data's short time frame (January to June 2023).

5. Conclusion:

Using the sales data given and the assumed cost structure, the analysis was able to determine which products were the most and least profitable. Clear insights into the sales and profit distribution were made possible by the visualizations.

Important findings include:

- Determining the ten most and least profitable products.
- Sales and profit for the top 20 products are visualized, showing how these two metrics relate to one another.

- Examination of possible causes of low profitability for the least profitable products, including unit price in relation to cost and sales volume.

Following suggestions are offered considering these findings:

Concentrate on High-Profit Products: To optimize sales and profit, give top priority to marketing and sales initiatives for the best-performing products.

Examine and Deal with Low-Profit Items: Examine the causes of some products' poor profitability. If profitability cannot be increased, this may entail a more thorough cost analysis, price changes, focused marketing, or the possibility of discontinuation.

Examine Time Series Forecasting in More Detail:

Although difficult with the available data, more time series modeling research may yield insightful information about future profit trends and seasonality for improved planning and decision-making.

Perform a thorough cost analysis: To calculate profit margins precisely and find areas for cost reduction, a more thorough grasp of the costs of individual products is essential.

Boost Operational Efficiency: To cut costs and boost overall profitability, always seek methods to improve operational procedures.